Nome _____ Cognome _____ Data ___/___/_____

Thermal physics: test, version A, answers and marking scheme

Total: 30 marks (A 12, B 14, C 4). Parts A and C: 1 mark for each correct answer. Part B: 2 marks for each of questions 13 to 16, 3 marks for each of questions 17 and 18, awarded with the grid printed under each question.

Answer key, multiple choice

1 B	**2** D	**3** A	**4** C	**5** B	**6** D
7 D	**8** C	**9** A	**10** B	**11** A	**12** D
19 D	**20** A	**22** C			

Written answers: **13** 2.3 mm **14** 315 s **15** 345 s **16** 390 J/(kg °C) **17** 44 °C **18** 30 °C **21** 22 min

A Multiple choice [12]

1. Answer B. $\Delta L = \alpha L_0 \Delta T$: L_0 and ΔT are the same, so the larger ΔL means the larger α.

A The specific heat capacity concerns the energy needed per kilogram and per degree, not the expansion. **C** Nothing in the question says so; both are heated by the same temperature change. **D** The question states that the two rods have the same length.

2. Answer D. Temperature measures the average kinetic energy of the particles, not the amount of water: same temperature, same average kinetic energy.

A There is more energy in total in Q, but it is shared among three times as many particles: the average per particle is the same. **B** The two beakers do not hold the same total energy: Q holds three times as much. What is equal is the average per particle. **C** More energy is needed to bring more water to 60 °C, but this energy is spread over more particles: each one ends up with the same average.

3. Answer A. A higher temperature means a higher average speed: more frequent and harder collisions with the walls.

B Particles do not change size. **C** The container is sealed: the number of particles is fixed. **D** The container is rigid, so the volume and the average spacing stay the same.

4. Answer C. Condensation gives back the energy that was absorbed in vaporisation ($Q = mL_v$), on top of the energy released as the water cools.

A The particles are the same: only their arrangement and energy change. **B** Both are at 100 °C, as stated in the question. **D** Density has nothing to do with the energy transferred to the skin.

5. Answer B. Only the fastest particles can leave the surface. They take more than the average energy with them, so the average of the others falls.

A They are not chosen at random: the particles that escape are the most energetic ones, so the average of those left behind falls. **C** Temperature does not depend on the amount of liquid. It depends on the average energy of the particles. **D** The liquid itself cools: the energy that makes the particles escape is taken from the liquid.

6. Answer D. Between R and S all the substance is liquid and the temperature rises.

A Here the substance is a solid being warmed. **B** Here the solid melts: solid and liquid are present and the temperature is constant. **C** Here the liquid boils: liquid and gas are present and the temperature is constant.

7. Answer D. Poor conduction and no circulation: both of the main ways energy would leave through the air are blocked.

A Air is a poor conductor, which is why it works as an insulator. **B** Air absorbs very little radiation; the effect comes from conduction and convection. **C** Air at body temperature cannot warm the body; it only slows the loss of energy.

8. Answer C. Convection: the density difference makes the fluid circulate, so the element heats the whole volume.

A Heating a fluid makes it less dense, not denser. **B** Water is a poor conductor; the circulation is what spreads the energy. **D** Most of the energy is carried by the moving water, not by radiation.

9. Answer A. A shiny surface emits little infrared radiation and reflects radiation back: the loss by radiation, one of the main ways the potato loses energy, is reduced.

B Spreading the energy evenly does not reduce the total energy lost to the room. **C** A shiny surface reflects infrared radiation instead of absorbing it, and the potato is hotter than the room, so the net energy flows outwards. **D** Metals are good conductors whatever their thickness: the foil works by reducing radiation, not by blocking conduction.

10. Answer B. While ice and water are both present, the energy goes into melting the ice and the temperature stays at the melting point.

A The plain average is not valid: part of the energy is used to melt the ice. **C** Ice and water together in equilibrium are at 0 °C, not below. **D** The water has lost energy to the ice, so it has cooled.

11. Answer A. A larger k means a steeper straight line on the $\ln$ graph, that is a faster fall of $T - T_{\text{room}}$.

B Both end at room temperature, which is the same. **C** The opposite: a smaller k means a slower fall. **D** The room is the same, but k is different, so the rates are different.

12. Answer D. Random errors have both signs and partly cancel in an average.

A A zero error shifts every reading by the same amount, so averaging does not remove it. **B** A systematic error is the same in every reading, so averaging cannot remove it. **C** Averaging does reduce the random error.

B Short answers **[14]**

Marking rule, Part B. Questions 13 to 16: **2 marks** for the correct final answer (the equivalent forms listed are accepted); **1 mark** for the single condition listed under 1 mark; **0 marks** for anything else. Questions 17 and 18: **1 mark for each of the three lines**, each judged only on its own line: a mistake in one line is not penalised again in the next. There are no marks for a method that is not on the line or in the condition listed.

13. Answer: 2.3 mm

$\Delta L = \alpha L_0 \Delta T = 23 \times 10^{-6} \times 2.0 \times 50 = 2.3 \times 10^{-3}\ \text{m} = 2.3\ \text{mm}.$

2 marks	2.3 mm
1 mark	Correct apart from a power of ten (for example 2300 mm or 0.23 mm).
0 marks	Anything else.

Common wrong answers. 3.2 mm: used the final temperature (70) instead of the change (50). 0.0023 mm: wrote the answer in metres but labelled it millimetres.

14. Answer: 315 s

$E = mc\,\Delta T = 1.5 \times 4200 \times 40 = 2.52 \times 10^5\ \text{J}; \quad t = E/P = 2.52 \times 10^5 / 800 = 315\ \text{s}.$

2 marks	315 s (accept 3.2×10^2 s; 5.3 min if the unit is written)
1 mark	$E = 2.5 \times 10^5$ J (or 252 000 J) written on the page, whatever the final answer.
0 marks	Anything else.

Common wrong answers. 2.5×10^5 s: that is the energy; forgot to divide by the power. 433 s: used the final temperature (55) instead of the change (40).

15. Answer: 345 s

$E = mL_v = 0.15 \times 2.3 \times 10^6 = 3.45 \times 10^5\ \text{J}; \quad t = E/P = 3.45 \times 10^5 / 1000 = 345\ \text{s}.$

2 marks	345 s (accept 3.5×10^2 s; 5.8 min if the unit is written)
1 mark	$E = 3.5 \times 10^5$ J (or 345 000 J) written on the page, whatever the final answer.
0 marks	Anything else.

Common wrong answers. 3.5×10^5 s: that is the energy; forgot to divide by the power. 50 s: used the latent heat of fusion instead of vaporisation.

16. Answer: 390 J/(kg ℃)

$c = \dfrac{Q}{m\,\Delta T} = \dfrac{7800}{0.50 \times 40} = 390\ J/(kg\ ℃)$.

2 marks	390 J/(kg ℃)
1 mark	Correct apart from a power of ten (for example 0.39 or 390 000), or $\Delta T = 40$ ℃ written on the page, whatever the final answer.
0 marks	Anything else.

Common wrong answers. 260: used the final temperature (60) instead of the change (40). 195: forgot the mass.

17. Answer: 44 ℃

Energy lost = energy gained: $0.40\,(80 - T) = 0.60\,(T - 20)$, so $32 - 0.4T = 0.6T - 12$, $T = 44$ ℃.

Line 1	1 mark: The energy balance in symbols, $m_1 c\,(T_1 - T) = m_2 c\,(T - T_2)$ (or $T = \dfrac{m_1 T_1 + m_2 T_2}{m_1 + m_2}$).
Line 2	1 mark: The numbers substituted: $0.40\,(80 - T) = 0.60\,(T - 20)$ (or $\dfrac{0.40 \times 80 + 0.60 \times 20}{1.00}$).
Line 3	1 mark: 44 ℃, with the unit.

Common wrong answers. 50 ℃: took the plain average of 80 and 20 (valid only for equal masses). 56 ℃: swapped the two masses.

18. Answer: 30 ℃

$\Delta T = \dfrac{\Delta L}{\alpha L_0} = \dfrac{1.53 \times 10^{-3}}{17 \times 10^{-6} \times 3.0} = 30$ ℃.

Line 1	1 mark: The equation rearranged for ΔT: $\Delta T = \Delta L/(\alpha L_0)$. ($\Delta L = \alpha L_0 \Delta T$ alone is not enough.)
Line 2	1 mark: The numbers substituted, with ΔL in metres: $\dfrac{1.53 \times 10^{-3}}{17 \times 10^{-6} \times 3.0}$.
Line 3	1 mark: 30 ℃, with the unit.

Common wrong answers. 30 000 ℃: used 1.53 mm without converting to metres. 90 ℃: forgot the length of the rod (L_0).

C Alternative to practical [4]

19. Answer D. The line goes from 4.00 at $t = 0$ to 3.28 at $t = 1200$ s: gradient $= -0.72/1200 =$ –6.0 × 10^{-4}/s, so $k = 6.0 \times 10^{-4}$/s.

A The gradient is $-k$. The constant k itself is positive. **B** Used the change in y without dividing by the change in t. **C** That is $1/k$, a time (the time constant), not k.

20. Answer A. At 50.0 ℃, $T - T_{\text{room}} = 30.0$ and $\ln 30.0 = 3.40$. The line is at 3.40 when $t \approx 1000$ s.

B Used $\ln 50.0 = 3.91$: forgot to subtract the room temperature before taking the logarithm. **C** That is the last time in the graph, not the time at 50.0 ℃. **D** That is $1/k$, the time constant, not a time on the graph.

21. Answer: 22 min

Cup Y cooled along the same curve as cup X, because the cups and the room are the same. Cup Y is at 40.0 ℃ now, so read the time at which cup X is at 40 ℃: about 22 min. (Check: $20 + 60\,e^{-t/20} = 40$ gives $t = 20 \ln 3 = 22.0$ min.)

1 mark	22 min (accept any value from 21 min to 23 min)
0 marks	Anything else.

Common wrong answers. 40 min: used the temperature 40.0 ℃ as a time. 14 min: read the curve at 50 ℃. –22 min: sign error, the cup was filled before $t = 0$.

22. Answer C. The rate of cooling depends on the difference $T - T_{\text{room}}$, and at $t = 0$ cup X has the larger difference.

A The time already spent cooling does not matter: only the difference $T - T_{\text{room}}$ at that moment does. **B** The opposite: when the two temperatures are close, the transfer is slower. **D** The cups and the room are the same, but the temperatures are not: the difference $T - T_{\text{room}}$ is different.

Topics, for the analysis of the results.

Thermal expansion	1, 13, 18
Temperature and particles, gas pressure	2, 3
Changes of state, latent heat, evaporation	4, 5, 6, 15
Conduction, convection and radiation	7, 8, 9
Specific heat capacity and power	14, 16
Mixtures, including ice and water	10, 17
Cooling and the law of cooling	11, 19, 20, 21, 22
Laboratory skills: random and systematic errors, reading and interpreting graphs	12, 19, 20, 21, 22